

PC 602 CS	COMPUTER NETWORKS				
Prerequisites	Data Structures and Programming Concepts	L	T	P	C
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Evaluation	CIE	40 Marks	SEE		60 Marks

Course Objectives	
1	To study the design issues in network layer and various routing algorithms
2	To introduce internet routing architecture and protocols
3	To learn the flow control and congestion control algorithms in Transport Layer
4	To introduce the TCP/IP suite of protocols and the networked applications supported by it
5	To learn basic and advanced socket system calls

Course Outcomes	
On completion of this course, the student will be able to	
CO1	Apply the function of each layer of OSI and trace the flow of information from one node to another node in the network
CO2	Understand the principles of IP addressing and internet routing
CO3	Analyze the working of networked applications such as DNS, mail file transfer and www
CO4	Implement client-server socket-based networked applications.

UNIT – I
Data Communications: Components, analog and digital signals and Encoders, Modems, RS232 Interfacing. Switching: Circuit Switching, Message Switching and Packet Switching. Topologies – Concept of layering.-Protocols and Standards – ISO / OSI model, TCP/IP.

UNIT – II
Data Link Layer: Error Control: Error detection and correction (CRC and Hamming code for singlebit correction) Flow Control: stop and wait – - sliding window protocols-go Back-N ARQ – selective repeat ARQ MAC LAYER: Ethernet IEEE 802.3LAN, Manchester encoding, Binary exponential algorithm, Efficiency calculation, ARP and RARP.

UNIT – III
Network Layer : Internetworks – virtual circuit and Datagram approach Routing – Distance Vector Routing ,Link State Routing , OSPF and BGP IPv4 , addressing, Subnetting, IPv6, CIDR, ICMP and IGMP protocols.

UNIT – IV
Transport Layer: Services of transport layer, Multiplexing and crash recovery Transmission Control Protocol (TCP) – TCP window management Congestion Control, timer management and User Datagram Protocol (UDP).

UNIT – V

Socket Programming: Primitive and advanced system calls, client/server iterative and concurrent programs IO multiplexing, Asynchronous IO and select system call.

APPLICATION LAYER : Domain Name Space (DNS) – SMTP – FTP – HTTP.

Suggested Reading:

1	<i>Computer Networks</i> , 5 th Edition, Andrew S. Tanenbaum , David J. Wetherall , Pearson Education, 2021
2	<i>Computer Networks: A Systems Approach</i> , Larry Peterson and Bruce Davie, Elsevier , 5 th Edition, 2021
3	<i>Computer Networking: A Top-Down Approach</i> , 6 th Edition, James F. Kurose , Keith W. Ross , Pearson , 2022